

# Lithium-Ion Batteries

A **Lithium-ion battery** is a rechargeable electrochemical energy storage device where lithium ions move between the **anode (usually graphite)** and **cathode (lithium metal oxide)** during charge/discharge.

- **Charging:**  $\text{Li}^+$  ions move from cathode  $\rightarrow$  anode
- **Discharging:**  $\text{Li}^+$  ions move from anode  $\rightarrow$  cathode

## Properties of Lithium-Ion Batteries

$\text{LiFePO}_4$  (Lithium Iron Phosphate)

Longer life, Higher usable capacity, Better efficiency

- ★ High DoD (Depth of Discharge) - 90%
- ★ Number of charging/discharging cycles
- ★ Stable chemistry (safe for rural deployment)
- ★ Long cycle life (2000–6000+ cycles)
- ★ High usable capacity (80–90% DoD)
- ★ Works well with BMS + data collection
- ★ Industry standard for **microgrids & solar storage**

Nominal Voltage:

- 12.8V (single module) (4S)
- 48V systems common (51.2V nominal) (standard for solar microgrids) (15S)

Capacity:

- 50 Ah – 200 Ah (modular)
- 5 kWh, 10 kWh packs common

## Lithium Iron Phosphate (LFP / $\text{LiFePO}_4$ )

Key Properties:

- Voltage:  $\sim 3.2\text{V/cell}$
- Energy density: 90–160 Wh/kg
- Cycle life: **2000–6000+ cycles**

- Thermal stability: Very high (safe)
- Cost: Moderate

### Parameters:

- Depth of Discharge (DoD): up to 90–100%
- Efficiency: ~95–98%
- Operating temp: -20°C to 60°C

## Key Parameters

| Parameter                       | Meaning                           | Typical Value (LFP) |
|---------------------------------|-----------------------------------|---------------------|
| <b>Energy Density (Wh/kg)</b>   | How much energy per weight        | 90–160 Wh/kg        |
| <b>Power Density (W/kg)</b>     | How fast energy can be delivered  | 250–1000 W/kg       |
| <b>Cycle Life</b>               | Number of charge/discharge cycles | 2000–6000+ cycles   |
| <b>Depth of Discharge (DoD)</b> | Usable capacity (%)               | 80–100%             |
| <b>State of Charge (SoC)</b>    | Current charge level              | 0–100%              |
| <b>State of Health (SoH)</b>    | Battery aging condition           | >80% (usable)       |
| <b>Efficiency (%)</b>           | Charge-discharge efficiency       | 95–98%              |
| <b>C-rate</b>                   | Charge/discharge speed            | 0.5C–1C typical     |

|                          |                           |               |
|--------------------------|---------------------------|---------------|
| <b>Thermal Stability</b> | Safety under heat         | Very high     |
| <b>Nominal Voltage</b>   | Average operating voltage | 3.2V per cell |

# Key Battery Parameters

Assume: 12.8V LFP battery, 100Ah

| Parameter                          | Explanation                     | Ideal Value (LFP 12V)                  |
|------------------------------------|---------------------------------|----------------------------------------|
| <b>Voltage</b>                     | Electrical potential of battery | 12.8V nominal (10–14.6V range)         |
| <b>Current</b>                     | Flow of charge                  | Depends on load (e.g., 10–50A typical) |
| <b>Energy</b>                      | Total stored energy             | ~1280 Wh (12.8V × 100Ah)               |
| <b>Power</b>                       | Rate of energy usage            | Depends on load (e.g., 500W–1kW)       |
| <b>Temperature</b>                 | Operating condition             | 15°C – 35°C ideal                      |
| <b>SoC</b>                         | Charge level (%)                | Operate between 20–90%                 |
| <b>SoH</b>                         | Battery health                  | >90% good, <80% aging                  |
| <b>DoD</b>                         | % of capacity used              | Up to 80–90% daily                     |
| <b>Lifetime</b>                    | Total usable years              | 8–15 years                             |
| <b>C-rating</b>                    | Charge/discharge rate           | 0.5C ideal (50A for 100Ah)             |
| <b>Cycle Counting</b>              | Number of cycles used           | Track up to ~3000+                     |
| <b>RUL (Remaining Useful Life)</b> | Time before failure             | Estimated via SoH + cycles             |
| <b>Internal Resistance</b>         | Opposition to current flow      | Very low (~2–10 mΩ)                    |